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A First Course in Contemporary Discrete Mathematics

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1

Foundations

We assume familiarity with elementary set-theoretic notation. The axioms of set theory are regarded as the foundational axioms of mathematics. There are many axiomatizations of set theory, but most common and widely accepted consists of the Zermelo–Fraenkel axioms together with the axiom of choice. This system is known as ZFC (Zermelo-Fraenkel plus Choice). One of the most elementary axioms of Zermelo-Fraenkel is that if a and b are given sets, then we can form the set $\{a, b\}$ whose elements are exactly a and b. Note that $\{a, b\} = \{b, a\}$, and if $a = b$, then the set $\{a, b\}$ contains only one element: a; this one-element set is denoted $\{a\}$, and it is often called a *singleton*. A set of the form $\{a, b\}$ is called an *unordered pair* (or, less commonly, a *doubleton*). In mathematics, we also need the important notion of *ordered pair*. In Zermelo-Fraenkel set theory, the ordered pair (a, b) is defined as the set $\{\{a\}, \{a, b\}\}$. Note that (a, b) is not equal to (b, a) (unless, of course, a, b). Similarly, we define ordered triples, quadruples, quintuples, and, in genral n tupkles $(a_1, \ldots, a_n)$ for any nonnegative integer n. If A is a set, the set of n-tuples $(a_1, \ldots, a_n)$ with $a_1, \ldots, a_n \in A$ is denoted A^n.

1.1 Relations

Definition 1.1. Let A be a set and n a nonnegative integer. An *n-ary relation* on A is a function $R\colon A^n \to \{0, 1\}$.

Thus, a relation R on A takes n-tuples of elements of A as input, and for each such input, outputs a value either 1 (intuitively meaning "True") or 0 (intuitively meaning "False"). Later in the course, we will consider relations that take values more complex than the simple set $\{0, 1\}$ of "truth-values."

Notation 1.2. Clearly, a relation $R\colon A^n \to \{0, 1\}$ is completely determined by its *truth set*

$$R^{-1}(1) = \{\, (a_1, \ldots, a_n) \in A^n \colon R(a_1, \ldots, a_n) = 1 \,\}.$$

Hence, we may (and will) identify R with this set. Thus, if $(a_1, \ldots, a_n) \in A^n$, we will write

$$(a_1, \ldots, a_n) \in R$$

in place of

$$R(a_1, \ldots, a_n) = 1.$$

Notation 1.3. If R is a binary relation on a set A and $a, b \in A$, we will write

$$a \; R \; b$$

as a shorthand for

$$R(a, b) = 1.$$

Similarly, we write $a \not R \; b$ as a shorthand for $\neg(a \; R \; b)$.

Note that this convenient notation is only possible for binary relations.

Definition 1.4. Let R be a binary relation on a set A.

(1) We say that R is *reflexive* if

$$\text{for all } a \in A, \text{ we have } a \; R \; a.$$

(2) We say that R is *antireflexive* if

$$\text{for all } a \in A, \text{ we have } \neg(a \; R \; a).$$

(3) We say that R is *symmetric* if

$$\text{for all } a, b \in A, \text{ whenever } a \; R \; b, \text{ we have } b \; R \; a.$$

(4) We say that R is *asymmetric* if

$$\text{for all } a, b \in A, \text{ whenever } a \; R \; b, \text{ we have } \neg(b \; R \; a).$$

(5) We say that R is *antisymmetric* if

$$\text{for all } a, b \in A, \text{ whenever } a \; R \; b \text{ and } b \; R \; a, \text{ we have } a = b.$$

(6) We say that R is *transitive* if

$$\text{for all } a, b, c \in A, \text{ whenever } a \; R \; b \text{ and } b \; R \; c, \text{ we also have } a \; R \; c.$$

Definition 1.5. Let A be a set.

(1) A *partial order on A* is a binary relation on A that is reflexive, antisymmetric, and transitive. We say that a partial order on A is *total* (or *linear*) if

$$\text{for all } a, b \in A, \text{ we have } a \; R \; b \text{ or } b \; R \; a.$$

If R is a partial order on A, then the pair (A, R) is called a *partially ordered* set. If R is total (or linear), then we call the pair (A, R) a *linearly ordered* set.

(2) Two partially ordered sets $(A, \leq_A)$ and $(B, \leq_B)$ are said to be *isomorphic* if there exists a bijection $f : A \to B$ such that for all $a, a' \in A$,

$$a \leq_A a' \quad \text{if and only if} \quad f(a) \leq_B f(a').$$

(3) A *strict order on A* is a binary relation on A that is antireflexive, asymmetric, and transitive.

Exercise 1.6. Let A be a set.

(1) Prove that if $\leq$ is a partial order on A and we define a new binary relation $<$ on A by

$$a < b \text{ if and only if } a \leq b \text{ and } a \neq b,$$

then $<$ is a strict order on A.

(2) Prove that if $<$ is a strict order on A and we define a new binary relation $\leq$ on A by

$$a \leq b \text{ if and only if } a < b \text{ or } a = b,$$

then $\leq$ is a partial order on A.

Exercise 1.7. For each of the following properties, exhibit a relation R that satisfies one of the three properties but not the other two. When such an example is not possible, prove why.

(i) R is reflexive,

(ii) R is antisymmetric,

(iii) R is transitive.

In this section, we have discussed three representation of relations:

(1) The functional representation of Definition 3.13,

(2) The set representation discussed in 3.13,

(3) The graph representation.

Of these, the set representation is the traditional one found in beginning textbooks. The graph representation is visually useful, but it is limited to binary relations. As we shall see in Section 3.3, the functional representation is particularly adaptable to contemporary multivalued contexts.

1.2 Alphabets and Words

Recall that a set A is *countable* or *enumerable* if there exists a surjective function $f : \mathbb{N} \to A$. The surjectivity of f means that

$$A = \{f(0), f(1), f(2), \dots\}.$$

We call f an *enumeration* of A. A set is *uncountable* if it is not countable.

Proposition 1.8. *If*

$$A_0, A_1, \dots, A_n, \dots \qquad n \in \mathbb{N}$$

are countable sets, then the union

$$\bigcup_{n \in \mathbb{N}} A_n$$

is countable.

Proof. For each $n \in \mathbb{N}$, let $f_n : \mathbb{N} \to A_n$ be an enumeration of A_n. Thus,

$$\begin{aligned} A_0 =& \{f_0(0), f_0(1), f_0(2), \dots\} \\ A_1 =& \{f_1(0), f_1(1), f_1(2), \dots\} \\ A_2 =& \{f_2(0), f_2(1), f_2(2), \dots\} \\ &\vdots \end{aligned}$$

Define a function

$$g : \mathbb{N} \to \bigcup_{n \in \mathbb{N}} A_n$$

as follows. Start with $g(0) = f_0(0)$; then let $g(1), g(2)$ list all the elements of the form $f_i(j)$ such that $i + j = 1$; then let $g(3), g(4), g(5)$ list all the elements of the form $f_i(j)$ such that $i + j = 2$; and so on. Thus, the function g is clearly surjective. □

Proposition 1.9. *If A and B are countable sets, then $A \times B$ is countable.*

Proof. This is immediate from Proposition 1.8 since if f, g are enumerations of A and B, respectively, then

$$A \times B = \bigcup_{n \in \mathbb{N}} \{(f(m), g(n)) \colon m \in \mathbb{N}\}.$$

□

Corollary 1.10. *If $A_1, \dots, A_k$ are countable, then $A_1 \times \cdots \times A_k$ is countable.*

Proof. Immediate from the preceding proposition and induction. □

Fix a set A, and let's think of A as an alphabet (or set of symbols) that we will use to form words. Mathematically, for $n \in \mathbb{N}$, the words of length n formed with symbols out of the alphabet A are defined as the elements of the set

$$A^n = \underbrace{A \times \cdots \times A}_{n \text{ times}}.$$

Thus, if A is countable, then by Corollary 1.10, the number of words of length n formed with symbols out of the alphabet A is countable. Moreover, by Proposition 1.8, the number of words of arbitrary finite length formed with symbols from the alphabet A,

$$\bigcup_{n \in \mathbb{N}} A^n,$$

is countable as well.

We have proved the following important result:

Corollary 1.11. *If A is a countable alphabet, then the number of words of finite length formed with symbols out of A is countable.*

Now let us switch our attention to words of infinite length. If A is an alphabet, a word of countably infinite length formed from the alphabet A is defined as an element of the set

$$A^{\mathbb{N}} = \underbrace{A \times A \times \dots}_{\text{countably infinitely many times}}.$$

In sharp contrast with Corollary 1.11, we have the following theorem:

Theorem 1.12. *Let A be an alphabet with at least two symbols. Then, the set of words of infinite length formed from the alphabet A is uncountable.*

Proof. It suffices to consider the case when A consists of only two symbols, let's call them 0 and 1. We show that

$$\{0,1\}^{\mathbb{N}} = \underbrace{\{0,1\} \times \{0,1\} \times \dots}_{\text{countably many times}}.$$

is uncountable. Note that $\{0,1\}^{\mathbb{N}}$ consists of all the infinite sequences of 0's and 1's, i.e.,

$$\{0,1\}^{\mathbb{N}} = \{\, (a_n)_{n\in\mathbb{N}} \colon \text{For each } n \in \mathbb{N},\ a_n \in \{0,1\}\,\}. \tag{*}$$

Suppose, by way of contradiction, that $\{0,1\}^{\mathbb{N}}$ is countable, and let f be an enumeration of $\{0,1\}^{\mathbb{N}}$; i.e.,

$$\{0,1\}^{\mathbb{N}} = \{f(0), f(1), f(2), \dots\}.$$

By $(*)$, for each $m \in \mathbb{N}$ there exists a sequence $(a_n^m)_{n\in\mathbb{N}} \in \{0,1\}^{\mathbb{N}}$ such that

$$f(m) = (a_n^m)_{n\in\mathbb{N}}. \tag{**}$$

Define a sequence $(a_n^*)_{n\in\mathbb{N}} \in \{0,1\}^{\mathbb{N}}$ by

$$a_n^* = \begin{cases} 0, & \text{if } a_n^n = 1, \\ 1, & \text{if } a_n^n = 0. \end{cases}$$

Then the sequence $(a_n^*)_{n\in\mathbb{N}}$ is in $\{0,1\}^{\mathbb{N}}$, but by $(**)$, it is not of the form $f(m)$ for any $m \in \mathbb{N}$, contradicting the assumption that f is surjective.

□

Let us reflect on what the last two results (namely, Corollary 1.11 and Theorem 1.12) show together. Corollary 1.11 shows that the set of finite strings of 0's and 1's is countable, and Theorem 1.12 shows that the set of infinite strings of 0's and 1's is uncountable. A finite string of 0's and 1's can be thought of as a computer program, or as a rational number written in binary form. An infinite string of 0's and 1's can be thought of as an irrational number written in binary form. Thus, the set of computer programs and the set of all rational numbers are countable, while the set of irrational numbers is uncountable.

To summarize, the size of the set of irrational numbers is much larger than the size of the set of rational numbers. It is then natural to ask: What is the difference between the two different sizes? As we will discuss in Section 1.3.2, this question is the famous *Continuum Hypothesis* (the very first of David Hilbert's famous list of open problems). It arose in the 20th century with the set-theoretic formalization of mathematics, and today it is known that, stunningly, it cannot be answered by the standard axioms of mathematics.

Exercises 1.13.

(1) Prove the following version of the Pigeonhole Principle:

Fact 1 (Pigeonhole Principle (countable/uncountable version))**.** *If every pigeon must be given a pigeonhole, but the number of pigeonholes is countable and the* *number of pigeons is uncountable, then at least one pigeonhole corresponds to an uncountable number of pigeons.*

(2) Use Corollary 1.11 and Theorem 1.12 to determine which of the following sets are countable:

(a) The set of all colorings of K_n, for a fixed $n \in \mathbb{N}$ into two colors.
(b) The set of all colorings of K_n, for all $n \in \mathbb{N}$ into two colors.
(c) The set of all colorings of $K_{\mathbb{N}}$ into two colors.

(3) For $k \in \mathbb{N}$, let p_k denote the kth prime number (so $p_1 = 2$, $p_2 = 3$, $p_3 = 5$, etc). Every integer $n \geq 2$ can be written uniquely as

$$n = p_1^{e_1} \cdots p_k^{e_k},$$

where $e_1, \ldots, e_k$ are nonnegative integers, $p_k > 0$, and no prime larger than p_k divides n. Thus, n can be seen as encoding the sequence $(e_1, \ldots, e_k)$. Use this to give a *direct proof* of Corollary 1.11 without invoking any of the lemmas proved in this section.

1.3 Well-Ordered Sets, Transfinite Induction, and Ordinal and Cardinal Numbers

1.3.1 Well-Ordered Sets and Transfinite Induction

Definition 1.14. Let $(A, \leq)$ be a partially ordered set. If S is a subset of A, we say that S has a *least* or *minimum* if there exists $m \in A$ such that

$$m \in S \quad \text{and} \quad \forall x \in S\ (m \leq x).$$

Note that (by the antisymmetry of the relation $\leq$) if such an element m exists, it must be unique; we denote it as $\min(S)$.

We will say that $(A, \leq)$ is a *well-ordered set* if every nonempty subset of A has a least element under $\leq$.

Remarks 1.15. (1) Every well-ordered set is linearly ordered, but the converse is not true (e.g., the interval $(0, 1)$ is linearly ordered but not well-ordered).

(2) Every well-ordered set has a minimum element, but the converse is not true (e.g., the interval $[0, 1]$ is linearly ordered but not well-ordered).

Exercises 1.16.

(1) Given a partial order $(A, \leq)$ and an element $a \in A$, define the set of *predecessors of* a as

$$\operatorname{pred}(a) = \{\, x \in A \colon x < a \,\}.$$

Prove that $(A, \leq)$ is well-ordered if and only if $(\operatorname{pred}(a), \leq)$ is well-ordered for every $a \in A$.

(2) Let $(A, \leq)$ and $(B, \trianglelefteq)$ be well-ordered sets. Define an order $\preceq$ on $A \times B$ as follows: If $(a, b), (a', b') \in A \times B$, then

$$(a, b) \preceq (a', b') \quad \text{if and only if} \quad a < a', \text{ or } a = a' \text{ and } b \trianglelefteq b'.$$

Prove that $(A \times B, \preceq)$ is a well-ordered set.

Proposition 1.17. *A nonempty partially ordered set $(A, \leq)$ is well-ordered if and only if there is no infinite descending chain*

$$a_0 > a_1 > a_2 > \dots$$

in A.

Proof. Suppose that $(A, \leq)$ is well-ordered and A contains an infinite descending chain as above. Then the set $\{S_n : n \in \mathbb{N}\}$ is a nonempty subset of A with no smallest element, contradicting the well-order assumption on $(A, \leq)$.

Conversely, suppose that A is nonempty and $(A, \leq)$ is not well-ordered. Since A is nonempty, fix $a_0 \in A$. Since A has no smallest element, then there exists $a_1 < a_0$. Now, since A has no smallest element, the element a_1 cannot be the minimum of A either, so there must be $a_2 < a_1$ in A. Iterating this process recursively, we find an infinite descending chain $a_0 > a_1 > a_2 > \dots$ in A. □

What makes well-ordered sets important in mathematics is the following fundamental principle:

Theorem 1.18 (Principle of Transfinite Induction)**.** *Suppose that $(A, \leq)$ is well-ordered. Then, in order to prove that every element of A satisfies a given property P, it suffices to prove the following condition:*

$$\text{If } a \in A \text{ and every } x < a \text{ satisfies } P\text{, then } a \text{ satisfies } P. \qquad \text{(IH)}$$

Proof. By contrapositive, suppose that $(A, \leq)$ is well-ordered, (IH) holds, and not every element of A satisfies P. Then the set of elements of A not satisfying P is nonempty, so by well-ordering, it has has a minimum element; let's call this element a. By the minimality of a, every $x < a$ satisfies P. But a does not. Thus, (IH) is contradicted. □

1.3.2 Ordinal and Cardinal Numbers

The ordinals are standard examples of well-ordered sets. They can also be seen as "extended numbers."

The ordinals are defined recursively as follows. The smallest ordinal is 0, which is formally defined as the empty set. The next smallest ordinal is 1, which is defined as $\{0\}$. After 0 and 1 comes 2, which is defined as $\{0, 1\}$. In general, $\{n + 1\}$ is defined as $\{0, \dots, n\}$. Formally, we define

- $0 = \emptyset$
- $n + 1 = n \cup \{n\}$.

These are all finite. In fact, these are called the *finite ordinals.* They are well-ordered sets in a natural way.

Note that if m, n are finite ordinals, then

$$m < n \quad \text{if and only if} \quad m \in n.$$

In other words, the well-ordering relation is the membership relation $\in$.

There is no reason to stop at the finite ordinals. The first infinite ordinal is

$$\omega = \{0, 1, 2, \dots\}.$$

Note that ω is well-ordered by Exercise 1. Formally, ω is defined as the union of all the preceding ordinals.

So now we have a (well-ordered) chain of ordinal numbers:

$$0, 1, 2, \ldots, \omega.$$

However, there is no reason to stop at ω. We can mimic what we did with the finite ordinals and define

$$\omega + 1 = \omega \cup \{\omega\} = \{0, 1, 2, \ldots, \omega\}.$$

More general for any ordinal, α, we can define the *successor* of α as follows:

Definition 1.19. If α is an ordinal, the *successor* of α is

$$\alpha + 1 = \alpha \cup \{\alpha\}.$$

Notice that, by Exercise 1, $\alpha + 1$ is well-ordered if α is well-ordered. Thus we now have

$$0, 1, 2, \ldots, \omega, \omega + 1, \omega + 2, \omega + 3, \ldots$$

But again, there is no reason to stop here. We can now define

$$\omega + \omega = \bigcup_{n<\omega} \omega + n.$$

Note that $\omega + \omega$ is well-ordered by Exercise 1. So, again, we have a chain

$$0, 1, 2, \ldots, \omega, \omega + 1, \omega + 2, \omega + 3, \ldots \omega + \omega, \omega + \omega + 1, \omega + \omega + 2, \ldots,$$

and yes: We also have

$$\omega + \omega + \omega = \bigcup_{n<\omega} \omega + \omega + n.$$

All of these sets are well-ordered by the relation $\in$.

Note that the ordinals ω, $\omega + \omega$, $\omega + \omega + \omega$ do not have immediate predecessors. Ordinals with no immediate predecessor are called *limit ordinals*. The formal definition is as follows:

Definition 1.20. An ordinal α is said to be a *successor ordinal* if α has an immediate predecessor, i.e., there exists an ordinal $\beta < \alpha$ such that $\alpha = \beta + 1$. An ordinal that is not a successor ordinal is called a *limit ordinal*.

Note that the first limit ordinal is ω.

We haven't really given a formal definition of the concept of "ordinal." The formal definition is as follows:

Definition 1.21. (1) 0 is an ordinal;

(2) If α is an ordinal, then $\alpha + 1 = \alpha \cup \{\alpha\}$ is an ordinal;

(3) A union of any chain of ordinals is an ordinal.

Since ordinals are defined recursively, by Theorem 1.18, we can define ordinal addition, multiplication, and exponentiation reursively as follows.

Fix an ordinal α, and by induction on β we define $\alpha+\beta$, $\alpha\cdot\beta$, and α^β through the following definitions (note that the definition of addition is based on the definition of successor given above):

Ordinal addition

- $\alpha + 0 = \alpha$,

- $\alpha + (\beta + 1) = (\alpha + \beta) + 1$,
- If β is a limit ordinal,
$$\alpha + \beta = \bigcup_{\xi < \beta} \alpha + \xi.$$

Ordinal multiplication

- $\alpha \cdot 0 = 0$,
- $\alpha \cdot (\beta + 1) = (\alpha \cdot \beta) + \alpha$,
- If β is a limit ordinal,
$$\alpha \cdot \beta = \bigcup_{\xi < \beta} \alpha \cdot \xi.$$

Ordinal exponentiation

- $\alpha^0 = 1$,
- $\alpha^{\beta+1} = \alpha^\beta \cdot \alpha$,
- If β is a limit ordinal,
$$\alpha^\beta = \bigcup_{\xi < \beta} \alpha^\xi.$$

Exercise 1.22. Prove that ordinal addition and multiplication are not commutative; more precisely, exhibit

(1) Ordinals α, β such that $\alpha + \beta \neq \beta + \alpha$.

(2) Ordinals α, β such that $\alpha \cdot \beta \neq \beta \cdot \alpha$.

Recall that two sets A and B are said to have the same cardinality if there is a bijective function between them.

Note that, since a countable union of countable sets is countable, all of the ordinals
$$0, 1, 2, \ldots \omega, \omega + 1, \omega + 2, \ldots \omega + \omega, \ldots \omega + \omega + \omega, \ldots \omega^\omega, \ldots$$
are countable. However, the axioms of set theory (which fall outside the scope of this course) ensure that there is a least ordinal, called ω_1, such that there is no bijective correspondence between ω_1 and any of its predecessors. In other words, the cardinality of ω_1 is strictly larger than the cardinality of any ordinal $\alpha < \omega_1$. Similarly, there is an ordinal ω_2 of cardinality strictly larger than that of ω_1. In general, we define

- $\omega_0 = \omega$, and
- For any ordinal α,
$$\omega_\alpha = \text{the least ordinal of cardinality larger than the cardinality of } \omega_\beta, \text{ for all } \beta < \alpha.$$

In other words, the ordinals of the form ω_α are those where a “step up” in cardinality occurs. Not surprisingly, these cardinalities are special, so there is a particular name and nomenclature reserved for them:

Definition 1.23. For each ordinal α, we let

$$\aleph_\alpha = \text{The set } \omega_\alpha.$$

In other words, $\aleph_\alpha$ is a name we give to the ordinal ω_α after we strip it of its well-order relation and think of it not as a well-ordered set, but rather as a set - giving measure of size for infinite sets. If A is an arbitrary set and there exists a bijection between A and $\aleph_\alpha$ ($=\omega_\alpha$), we write

$$|A| = \aleph_\alpha.$$

If A, B are sets and $|A| = \aleph_\alpha = |B|$, we simply write $|A| = |B|$ and say that A and B are *equipotent.*

The following is one of the many equivalent forms of the famous Axiom of Choice from set theory:

Fact 2 (The Axiom of Choice (AC))**.** *For every infinite set A there exists an ordinal α such that $|A| = \aleph_\alpha$ i.e., the cardinality of A is $\aleph_\alpha$.*

Kurt Gödel (1937) and Paul J. Cohen (1963) proved that this axiom cannot be proved or disproved from the standard axioms of set theory.[1] So, it is regarded as a special axiom.

Now since $\aleph_\alpha$ is nothing but ω_α, and the latter is a well-ordered set, then Fact 2 says that any set whatsoever can be put into bijective correspondence with a well-ordered set. Therefore, every set can be well-ordered. This is the famous Well-Ordering Principle:

Fact 3 (The Well-Ordering Principle (WOP))**.** *For every set A there exists a binary relation $\leq$ on A such that $(A, \leq)$ is a well-ordered set.*

Remarks 1.24. (1) We have just observed that (AC) implies (WOP). The opposite implication is also true, because if A is any set and $\leq$ is a well-ordering relation on A, then by Exercise , $(A, \leq)$ is isomorphic to an ordinal, i.e., there exists an ordinal α such that $(A, \leq)$ is isomorphic to $(\alpha, <)$. But, then one can take the smallest such α, and such α by definition must be a cardinal. Thus, A is equipotent to a cardinal.

(2) A useful, and rather surprising, consequence of Fact 3 is that, by 1.18 do induction proofs on any set.

We know (by Theorem 1.12) that the real line, $\mathbb{R}$, is uncountable i.e., $|\mathbb{R}| > \aleph_0$. But then, what is $|\mathbb{R}|$? Is it $\aleph_1$? Or maybe $\aleph_2$? Or maybe something much bigger, say, $\aleph_\omega$? Georg Cantor developed the theory of ordinal and cardinal numbers in the late 19th century in order to settle this question (motivated by his investigations in analysis). He believed that $|\mathbb{R}| = \aleph_1$, i.e., the size of the real line is the smallest uncountable cardinality. This became known as Cantor's Continuum Hypothesis. Cantor spent the rest of his life unsuccessfully trying to prove it. His inability to devise a solution led to increasingly frequent bouts of depression, and he died in a sanatorium in 1918. However, his groundbreaking theories of transfinite ordinal

[1]Gödel proved that AC cannot be proved to be false from the axioms of set theory, and Cohen proved that that AC cannot be proved to be true from said axioms.

and cardinal numbers transformed mathematics profoundly (it was claimed that the number of theorems proved between 1900 and 1940 was larger than the number of theorems proved between Archimedes and 1900), and became a cornerstone of the mathematics of the new century. In the words of Hilbert, the most influential mathematician of the 20th century, Cantorian set theory brought mathematicians into "paradise." Naturally, Cantor's Continuum Hypothesis became the very first problem in Hilbert's famous list of open problems, which would shape the course of mathematics during the next 100 years and into our time.

The question is profound: What is the cardinality of the real line?

In 1937, Kurt Gödel constructed universes of mathematics where the Continuum Hypothesis is true (that is, the cardinality of the continuum is the smallest uncountable cardinality), and in 1963, Paul J. Cohen constructed universes of mathematics where it is false. Thus, Cantor's Continuum Hypothesis cannot be proved or disproved from the standard axioms of set theory. Cohen won the Fields Medal and the National Medal of Science for his achievement, but the true size of the set of real numbers remains one of the great mysteries of mathematics.

Exercise 1.25. Prove that for every well-ordered set $(A, \leq)$ there is an ordinal α such that $(A, <)$ is isomorphic to $(\alpha, \in)$. [*Hint*: Recall the definition of $\mathrm{pred}(a)$ from Exercise 1.16. Prove that for every $a \in A$ there exists an ordinal α_a and an isomorphism f_a between $(\mathrm{pred}(a), <)$ and $(\alpha, \in)$ such that for all $b, c \in \mathrm{pred}(a)$, $b < c \Rightarrow f_a(b) \subseteq f_a(c)$.]

2

Topology and limits

2.1 Topological Spaces

Definition 2.1. A *topological space* is a structure $(X, \mathscr{N}_x)_{x \in X}$ that consists of a set X and, for each point $x \in X$, a collection $\mathscr{N}_x$ of subsets of X that satisfy the following properties:

(1) $x \in U$ for every $U \in \mathscr{N}_x$;

(2) If $U, V \in \mathscr{N}_x$, then there exists $W \in \mathscr{N}_x$ such that $W \subseteq U \cap V$;

(3) For every $U \in \mathscr{N}_x$ and every $y \in U$, there exists $V_y \in \mathscr{N}_y$ such that $V_y \subseteq U$.

Next we introduce an immediate result from the previous definition. The reader should note that Property (3) and the next proposition are equivalent.

Proposition 2.2. *Let $(X, \mathscr{N}_x)_{x \in X}$ be a topological space. For every $U \in \mathscr{N}_x$ and every $y \in U$, there exists $V_y \in \mathscr{N}_y$ such that*

$$U = \bigcup_{y \in U} V_y.$$

Proof. Fix a point $x \in X$, and let $U \in \mathscr{N}_x$ be given. By Property (3) above, we have that for every $y \in U$, there is some $V_y \in \mathscr{N}_y$ such that $V_y \subseteq U$. Thus, the union of all such V_y is contained in U. Also, by Property (1), we see that for every $y \in U$, we have $y \in V_y \subseteq \bigcup_{y \in U} V_y$. Thus, $U = \bigcup_{y \in U} V_y$. □

Definition 2.3. Let $(X, \mathscr{N}_x)_{x \in X}$ be a topological space.

For each $x \in X$, we refer to $\mathscr{N}_x$ as the *neighborhood base* for x. The indexed family $(\mathscr{N}_x)_{x \in X}$ is called a *local neighborhood base* on X. For all $x \in X$, each set $U \in \mathscr{N}_x$ is called a *basic neighborhood* of x. A *neighborhood* of x is any superset of a basic neighborhood of x.

Definition 2.4. A set $O \subseteq X$ is called *open* if for every $x \in O$, there exists $V_x \in \mathscr{N}_x$ such that

$$O = \bigcup_{x \in O} V_x.$$

A set $C \subseteq X$ is called *closed* if $X \setminus C$ is open. Note that C is closed if for every $x \in X$, and every basic neighborhood of x that intersects C, then $x \in C$.

Note that, by Proposition 2.2, a subset of X is open if and only it is a union of basic neighborhoods.

Proposition 2.5. *In any topological space $(X, \mathscr{N}_x)_{x \in X}$, the following properties hold:*

(1) *Both $\emptyset$ and X are open subsets of X (and evidently also closed);*

(2) *Any union of open subsets of X is open;*

(3) *Any finite intersection of open subsets of X is open.*

Proof.

(1) The empty set is open since any condition that must be satisfied by the points in $\emptyset$ is true vacuously. The set X itself is open since, for each point $x \in X$, we may choose $V_x \in \mathscr{N}_x$ arbitrarily so that $X = \bigcup_{x \in X} V_x$.

(2) Given any open subsets $(O_i)_{i \in I}$ of X, we have that each O_i is a union of basic neighborhoods. Thus, the union $\bigcup_{i \in I} O_i$ is a larger union of basic neighborhoods, and hence also an open subset of X.

(3) If O_1 and O_2 are open subsets of X, then for each $x \in O_1 \cap O_2$, there are $U, V \in \mathscr{N}_x$ such that $U \subseteq O_1$ and $V \subseteq O_2$. By Property (2) of Definition 2.1, there's $W_x \in \mathscr{N}_x$ such that

$$W_x \subseteq U \cap V \subseteq O_1 \cap O_2.$$

Thus, $O_1 \cap O_2$ is the union of all such $W_x \in \mathscr{N}_x$ that satisfy $W_x \subseteq O_1 \cap O_2$ as x ranges over $O_1 \cap O_2$.

□

Note that an infinite intersection of open sets is not necessarily open, as will be shown in Example 2.7(a).

Definition 2.6. Let $(X, \mathscr{N}_x)_{x \in X}$ be a topological space.

The *topology* $\mathscr{T}$ on X generated by the local neighborhood base $(\mathscr{N}_x)_{x \in X}$ is the collection of all open subsets of X; that is,

$$\mathscr{T} = \{O \subseteq X : O \text{ is open}\}.$$

Note that two different local neighborhood bases on X can generate the same topology. For instance, examples (a) and (b) below generate the same topology on $\mathbb{R}$.

Example 2.7. Basic examples

(a) Let $X = \mathbb{R}$, and for each $x \in \mathbb{R}$, let $\mathscr{N}_x = \{(x-\varepsilon, x+\varepsilon) : \varepsilon > 0\}$.

We will show that there is an infinite intersection of open subsets of X that is not open. Take, for instance, the infinite intersection

$$\bigcap_{n\in\mathbb{N}} (0, 1+\frac{1}{n}) = (0,1].$$

This interval is not open since there is no basic neighborhood $(1-\varepsilon, 1+\varepsilon)$ of $1 \in X$ that is contained in $(0,1]$.

(b) Let $X = \mathbb{R}$, and for each $x \in \mathbb{R}$, let $\mathscr{N}_x = \{(\alpha,\beta)\colon \alpha < x < \beta, \alpha, \beta \in \mathbb{R}\}$.

(c) Let $X = \mathbb{R}^2$, and for each $x \in \mathbb{R}^2$, let $\mathscr{N}_x$ consist of the set of all open disks about x of radius ε for all $\varepsilon > 0$.

(d) Let X be any set, and for each $x \in X$, let $\mathscr{N}_x = \{\{x\}\}$; i.e., the only basic neighborhood of x is the singleton $\{x\}$. The resulting topology is called the *discrete* topology on X.

Note that, in this topological space, every subset of X is open (since every set is a union of singletons). Hence, every subset is closed as well.

(e) Let X be any set, and for each $x \in X$, let $\mathscr{N}_x = \{X\}$. The resulting topology is called the *trivial* (or *indiscrete*) topology on X. Note that, in this topological space, only two subsets are open; namely, $\emptyset$ and X.

Exercise 2.8. Let $(A, \leq)$ be a linearly ordered set (see Definition 1.5 and the exercises that follow it). For $a, b \in A$ with $a < b$, define the *open interval* (a,b) by

$$(a,b) = \{x \in A\colon a < x < b\}.$$

Now for $x \in A$, define

$$\mathscr{N}_x = \{\, (a,b)\colon a < x < b \,\}\,.$$

(a) Prove that $(A, \mathscr{N}_x)_{x\in A}$ is a topological space, i.e., show that $\mathscr{N}_x$ satisfies properties (1)–(3) of the definition of topological space, for all $x \in A$.(Definition 2.1).

(b) Suppose above that we replace "linearly ordered" by "partially ordered." Is $(A, \mathscr{N}_x)_{x\in A}$ a topological space? Why or why not?

(c) Suppose that we define

$$\mathscr{N}_x = \{\, [x,b)\colon x < b \,\}\,.$$

Is $(A, \mathscr{N}_x)_{x\in A}$ a topological space? Why or why not?

Exercises 2.9.

(1) For any given set A, let $\mathbb{R}^A$ denote the set of all functions from A into $\mathbb{R}$. We define a topology on $\mathbb{R}^A$. To this effect, fix $f \in \mathbb{R}^A$ in order to define a set $\mathscr{N}_f$ of basic neighborhoods of f.

Given $x_1, \dots, x_n \in A$ and $\varepsilon > 0$, define

$$V_f(x_1,\dots,x_n,\varepsilon) = \left\{g \in \mathbb{R}^A\colon |g(x_i) - f(x_i)| < \epsilon \text{ for all } i \in \{1,\dots,n\}\right\}.$$

Define now

$$\mathscr{N}_f = \{V_f(x_1,\dots,x_n,\varepsilon)\colon \varepsilon > 0, x_1,\dots,x_n \in A, n \in \mathbb{N}\}\,.$$

Prove that $(\mathbb{R}^A, \mathscr{N}_f)_{f\in\mathbb{R}^A}$ is a topological space, i.e., show that $\mathscr{N}_f$ satisfies properties (1)–(3) of the definition of topological space (Definition 2.1).

(2) Let $(X, \mathscr{N}_x)_{x\in X}$ be a topological space, let A be a set, and denote by X^A the set of all functions from A into X. Define a topology on X^A as follows. Given $f \in X^A$, $x_1, \ldots, x_n \in A$ and $V_1 \ldots, V_n \subseteq X$ such that $V_i \in \mathscr{N}_{f(x_i)}$ for $i \in \{1, \ldots, n\}$, define

$$V_f(x_1, \ldots, x_n, V_1, \ldots, V_n) = \left\{g \in X^A \colon g(x_i) \in V_{f(x_i)} \text{ for all } i \in \{1, \ldots, n\}\right\}.$$

Then define

$$\mathscr{N}_f = \left\{V_f(x_1, \ldots, x_n, V_1, \ldots, V_n) \colon V_i \in \mathscr{N}_{f(x_i)} \text{ for } i \in \{1, \ldots, n\}\right\}.$$

Prove that $(X^A, \mathscr{N}_f)_{f\in X^A}$ is a topological space, i.e., show that $\mathscr{N}_f$ satisfies properties (1)–(3) of the definition of topological space (Definition 2.1).

2.2 Topological Limits

We now transition to discuss the notion of limit in a topological space. Fix a topological space $(X, \mathscr{N}_x)_{x\in X}$ and an indexed family $(a_i)_{i\in I}$ in X. We want to express what it means for $(a_i)_{i\in I}$ to converge to a point $c \in X$. Intuitively, this means that $(a_i)_{i\in I}$ *clusters* around a point $c \in X$; that is, for every basic neighborhood $V_c \in \mathscr{N}_c$, there exists a "large" $L \subseteq I$ such that $\{a_i \colon i \in L\} \subseteq V_c$. To clarify what is meant by "large," we proceed by defining the notion of filter, which may be thought of as a collection of all the "large" subsets L of I.

Definition 2.10. Fix a nonempty set I. A *filter* on I is a collection $\mathscr{F}$ of subsets of I that satisfy the following properties:

(1) $\emptyset \notin \mathscr{F}$ and $I \in \mathscr{F}$;

(2) If $A \in \mathscr{F}$ and $B \supseteq A$, then $B \in \mathscr{F}$;

(3) If $A, B \in \mathscr{F}$, then $A \cap B \in \mathscr{F}$.

Similarly, we may define the notion of "small" in the following way.

Definition 2.11. Fix a nonempty set I. An *ideal* on I is a collection $\mathscr{J}$ of subsets of I that satisfy the following properties:

(1) $\emptyset \in \mathscr{J}$ and $I \notin \mathscr{J}$;

(2) If $A \in \mathscr{J}$ and $B \subseteq A$, then $B \in \mathscr{J}$;

(3) If $A, B \in \mathscr{J}$, then $A \cup B \in \mathscr{J}$.

Observe that given a filter $\mathscr{F}$ on a nonempty set I, we can define an ideal $\mathscr{J}$ on I by

$$\mathscr{J} = \{I \setminus A \colon A \in \mathscr{F}\}.$$

Similarly, given an ideal $\mathscr{J}$ on I, we see that

$$\mathscr{F} = \{I \setminus A \colon A \in \mathscr{J}\}$$

is a filter on I.

Notation 2.12. If $\mathscr{F}$ is a filter on a nonempty set I, then the elements of $\mathscr{F}$ are called *$\mathscr{F}$-large*. A property of elements of I is said to hold for *almost every* $i \in I$ or *almost everywhere* in I (mod $\mathscr{F}$) if the property holds for an $\mathscr{F}$-large subset of I.

We are now ready to define the notion of convergence.

Definition 2.13. Let $(X, \mathscr{N}_x)_{x\in X}$ be a topological space, and let $(a_i)_{i\in I}$ be an indexed family of elements in X. If $\mathscr{F}$ is a filter on I, we say a_i *converges* to a point $c \in X$ with respect to $\mathscr{F}$, denoted $a_i \overset{\mathscr{F}}{\to} c$, if and only if for every $V_c \in \mathscr{N}_c$, we have $\{i\colon a_i \in V_c\} \in \mathscr{F}$.

We call c an *$\mathscr{F}$-limit* of $(a_i)_{i\in I}$ if and only if $(a_i)_{i\in I}$ converges to c with respect to $\mathscr{F}$.

In the context of a sequence of real numbers, this is the definition we are familiar with. Consider Example 2.7(a); namely, the topological space $(\mathbb{R}, \mathscr{N}_x)_{x\in\mathbb{R}}$ where $\mathscr{N}_x = \{(x-\varepsilon, x+\varepsilon)\colon \varepsilon > 0\}$. Let $I = \mathbb{N}$, and let $\mathscr{F}$ consist of all cofinite subsets of $\mathbb{N}$. Then, if $(a_n)_{n\in\mathbb{N}}$ is a sequence of real numbers, we see that $a_n \overset{\mathscr{F}}{\to} c$ if and only if for every $V_c \in \mathscr{N}_c$, we have $a_n \in V_c$ for all but finitely many $n \in \mathbb{N}$; that is, for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n > N$, we have $a_n \in (c-\varepsilon, c+\varepsilon)$.

Definition 2.14. The filter that consists of all cofinite subsets of any nonempty set I is called the *Fréchet filter on I*.

Recall that, in the topological space $(\mathbb{R}, \mathscr{N}_x)_{x\in\mathbb{R}}$ mentioned above, limits are unique. However, this is not true in every topological space. Consider the indiscrete topology discussed in Example 2.7(e); namely, given any set X, the indiscrete topology on X is the set $\mathscr{T} = \{\emptyset, X\}$. Given any indexed family $(a_i)_{i\in I}$ in X and any filter $\mathscr{F}$ on I, since X is the only basic neighborhood of any point in X and

$$\{i\colon a_i \in X\} = I \in \mathscr{F},$$

it follows that every point in X is a limit of $(a_i)_{i\in I}$.

Thus, a new question arises: What property must a topological space have to ensure that every limit is unique? This question leads us to the following definition.

Definition 2.15. A topological space $(X, \mathscr{N}_x)_{x\in X}$ is called *Hausdorff* if any two distinct points in X can be separated by disjoint neighborhoods; i.e., if $x, y \in X$ such that $x \neq y$, then there exist basic neighborhoods $V_x \in \mathscr{N}_x$ and $V_y \in \mathscr{N}_y$ such that $V_x \cap V_y = \emptyset$.

Proposition 2.16. *Let $(X, \mathscr{N}_x)_{x\in X}$ be a Hausdorff topological space, let $(a_i)_{i\in I}$ be an indexed family in X, and let $\mathscr{F}$ be a filter on I. If $a_i \overset{\mathscr{F}}{\to} c$ and $a_i \overset{\mathscr{F}}{\to} d$, then $c = d$.*

Proof. Suppose, on the contrary, that

$$a_i \overset{\mathscr{F}}{\to} c,$$
$$a_i \overset{\mathscr{F}}{\to} d,$$

and $c \neq d$. Since $(X, \mathscr{N}_x)_{x\in X}$ is Hausdorff, we can separate c and d by disjoint neighborhoods, say, U and V. Then we have

$$A = \{i\colon a_i \in U\} \in \mathscr{F} \text{ and}$$
$$B = \{i\colon a_i \in V\} \in \mathscr{F},$$

so by Property (3) of Definition 2.10, we have $A \cap B = \emptyset \in \mathscr{F}$, which contradicts Property (1). We conclude $c = d$. $\square$

The converse of Proposition 2.16 is also true. Although it is not needed for the material in these notes, it can be proved easily, so we have included a proof in Proposition **??**

Notation 2.17. In case $\mathscr{F}$-limits are unique in a topological space $(X, \mathscr{N}_x)_{x\in X}$ and an indexed family $(a_i)_{i\in I}$ in X converges to $c \in X$, we write

$$c = \mathscr{F}\lim a_i \quad \text{or} \quad c = \lim_{i,\mathscr{F}} a_i.$$

2.2.1 Principal Ultrafilters

Definition 2.18. An *ultrafilter* $\mathscr{U}$ on a nonempty set I is a filter on I such that for every $A \subseteq I$, either $A \in \mathscr{U}$ or $I \setminus A \in \mathscr{U}$.

Filters can also be seen as generalized measures. An ultrafilter $\mathscr{U}$ on a set I divides the subsets of I into two categories: $\mathscr{U}$-large and $\mathscr{U}$-small — the large subsets of I are those in $\mathscr{U}$, and the small subsets of I are those whose complement is in $\mathscr{U}$. Thus, ultrafilters are "complete" measures in the sense that every set is measurable. Filters, in contrast, need not be complete in this sense: For an arbitrary filter $\mathscr{F}$ on a set I, there may be subsets of I that are neither $\mathscr{F}$-large nor $\mathscr{F}$-small (for instance, for the Fréchet filter on $\mathbb{N}$, any subset of $\mathbb{N}$ that is neither finite nor cofinite — e.g., the set of even numbers — is neither $\mathscr{F}$-large nor $\mathscr{F}$-small.)

Example 2.19. Principal Ultrafilters

Fix a nonempty set I and an element $a \in I$. Define

$$\mathscr{U}_a = \{A \subseteq I \colon a \in A\}.$$

Note that $\mathscr{U}_a$ is an ultrafilter on the set I. Any ultrafilter defined in this way is called a *principal* (or "trivial" or "fixed") ultrafilter. More precisely, $\mathscr{U}_a$ is called the principal ultrafilter on I *generated by* a.

The preceding example suggests that ultrafilters represent an abstract notion of "point." Principal ultrafilters are "hard" points, while nonprincipal ultrafilters are "soft" points.

Exercise 2.20. Prove that if I is finite, then every ultrafiter on I is principal.

2.2.2 Nonprincipal Ultrafilters

Proposition 2.21. *An ultrafilter $\mathscr{U}$ on a set I is principal if and only if $\mathscr{U}$ contains a finite subset of I.*

Proof. If $\mathscr{U}$ is principal, then $\mathscr{U}$ contains a finite set; namely, the singleton that generates it. Suppose, conversely, that $\mathscr{U}$ is nonprincipal. We prove, by induction on n, that $\mathscr{U}$ cannot contain n-element subsets of I for $n = 1, 2, \ldots$. The case $n = 1$ is given by the definition of nonprincipal. Suppose that $\mathscr{U}$ contains a set $\{a_1, \ldots, a_n\}$, where $a_1, \ldots, a_n$ are distinct elements of I. Since, by the non nonprincipality assumption, we cannot have $\{a_1\} \in \mathscr{U}$, we must have $I \setminus \{a_1\} \in \mathscr{U}$. But then, since $\mathscr{U}$ is closed under finite intersections, $\mathscr{U}$ must contain $(I \setminus \{a_1\}) \cap \{a_1, \ldots, a_n\} = \{a_2, \ldots, a_n\}$, which contradicts the induction assumption. $\square$

The preceding proposition can be re-stated as follows:

Proposition 2.22. *An ultrafilter $\mathscr{U}$ on an infinite set I is nonprincipal if and only if it extends the Fréchet filter on I.*

In light of the preceding proposition, to prove that nonprincipal ultrafilters exist, we only have to prove that the Fréchet filter on I can be extended to an ultrafilter. The following propositions show that, indeed, any collection of subsets of I that has the finite intersection property can be extended to an ultrafilter; see, in particular, Corollaries 2.26 and 2.27.

Definition 2.23. A collection $\mathscr{B}$ of subsets of I has the *finite intersection property* (or is *finitely consistent*) if whenever $A_1, \ldots, A_n \in \mathscr{B}$, we have

$$A_1 \cap \cdots \cap A_n \neq \emptyset.$$

The finite intersection property is ubiquitous in mathematics. Mathematicians often refer to it as "f.i.p."

Proposition 2.24. *Every collection of subsets of I that has the finite intersection property can be extended to a filter on I.*

Proof. Let $\mathscr{B}$ be a collection of subsets of I with the finite intersection property. Without loss of generality, we can assume that $\mathscr{B}$ is closed under finite intersections. But then, once we have assumed this, to extend $\mathscr{B}$ to a filter, we only have to close it under supersets; in other words, the collection

$$\mathscr{F} = \{A \subseteq I \colon A \supseteq B \text{ for some } B \in \mathscr{B}\}$$

is clearly a filter on I. □

Due to the preceding proposition, any collection $\mathscr{B}$ of subsets of I with the finite intersection property is often called a *base* for a filter (or a *filter base*) on I. The filter $\mathscr{F}$ constructed in the proof of the Proposition 2.24 is called the filter *generated* by $\mathscr{B}$.

Now we want to improve Proposition 2.24 by showing that every filter can be extended to an ultrafilter on I.

Proposition 2.25. *Let $\mathscr{F}$ be a filter on a set I. Then, for every $A \subseteq I$, there exists a filter $\mathscr{F}' \supseteq \mathscr{F}$ on I such that either $A \in \mathscr{F}'$ or $I \setminus A \in \mathscr{F}'$*

Proof. Fix a set $A \subseteq I$. By Proposition 2.24, it suffices to verify that either $\mathscr{F} \cup \{A\}$ or $\mathscr{F} \cup \{I \setminus A\}$ has the finite intersection property. But this is immediate because if $\mathscr{F} \cup \{A\}$ fails to have the finite intersection property, it is because there exist $A_1, \ldots, A_n \in \mathscr{F}$ such that $A_1 \cap \cdots \cap A_n \cap A = \emptyset$, and this means that $I \setminus A \supseteq A_1 \cap \cdots \cap A_n$, which implies that $\mathscr{F} \cup \{I \setminus A\}$ has the finite intersection property. □

Corollary 2.26 (The Ultrafilter Theorem [A. Tarski, 1930])**.** *Every filter on a set I can be extended to an ultrafilter on I.*

Proof. Fix a filter $\mathscr{F}$ on I, let $\alpha = |\mathscr{P}(I)|$ and let $(A_\xi)_{\xi<\alpha}$ be a list of all the subsets of I. By induction, we construct an increasing chain

$$\mathscr{F} = \mathscr{F}_0 \subseteq \mathscr{F}_1 \subseteq \cdots \subseteq \mathscr{F}_\xi \subseteq \ldots \qquad (\xi \leq \alpha)$$

of filters in I such that:

(1) If $\xi = \zeta + 1$, then
$$A_\zeta \in \mathscr{F}_\xi \quad \text{or} \quad I \setminus A_\zeta \in \mathscr{F}_\xi;$$

(2) If $\xi \leq \alpha$ is a nonzero limit ordinal, then
$$\mathscr{F}_\xi = \bigcup_{\zeta<\xi} \mathscr{F}_\zeta.$$

The construction is as follows. Fix $\xi < \alpha$ and that $\mathscr{F}_\zeta$ has been defined for all $\zeta < \xi$.

If ξ is a successor ordinal, say $\xi = \zeta + 1$, by Proposition 2.25, either $\mathscr{F}_\zeta \cup \{A_\zeta\}$ or $\mathscr{F}_\zeta \cup \{I \setminus A_\zeta\}$ has the finite intersection property (it is possible that both sets have the f.i.p.). We pick one of these sets that has the finite intersection property, and let $\mathscr{F}_\xi$ be a filter extending it.

If ξ is a limit ordinal, we simply let $\mathscr{F}_\xi = \bigcup_{\zeta<\xi} \mathscr{F}_\zeta$. Since $(\mathscr{F}_\zeta)_{\zeta<\xi}$ is an increasing chain of filters, it is easy to see that $\mathscr{F}_\xi$ is closed under finite intersections and supersets (for if $A, B \in \mathscr{F}_\xi$, then A and B are in some member of the chain), and does not contain the empty set, i.e., $\mathscr{F}_\xi$ is a filter.

By construction, $\mathscr{F}_\alpha$ is an ultrafilter containing $\mathscr{F}_0 = \mathscr{F}$.

□

Corollary 2.27. *Nonprincipal ultrafilters exist on any infinite set.*

Proof. Let I be an infinite set. By Proposition 2.25, the Fréchet filter on I can be extended to an ultrafilter. By Proposition 2.22, such an ultrafilter must be nonprincipal. □

Corollary 2.28. *Let $\mathscr{B}$ be a family of subsets of I with the finite intersection property. Then there exists an ultrafilter $\mathscr{U}$ on I such that every set in $\mathscr{B}$ is $\mathscr{U}$-large.*

Proof. By Proposition 2.24 and Corollary 2.26. □

3

Finding Patterns in Large Networks (or Why Complete Chaos Is Impossible)

3.1 Ramsey's Theorem (The Classical Version)

The following is regarded as the most important principle of discrete mathematics.

Theorem 3.1 (Pigeonhole Principle (general form))**.** *If A, B are sets such that the cardinality of A is strictly greater than the cardinality of B, then no function from A into B can be injective.*

This principle is proved formally using the axioms of set theory, so we will not prove it here, and instead, we shall take it as a given fact. It should be intuitively clear.

The pigeonhole principle is informally stated in terms of pigeons and pigeonholes: If every pigeon must be given a pigeonhole, but there are more pigeons than pigeonholes, then at least one pigeonhole corresponds to more than one pigeon.

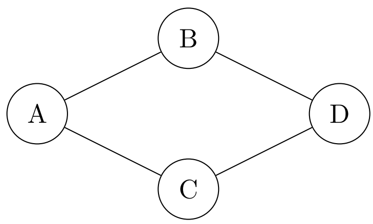
An immediate corollary of Theorem 3.1 is the following:

Theorem 3.2 (Pigeonhole Principle (infinite form))**.** *If A is infinite and B is finite, then there is $b \in B$ such that $f^{-1}(b)$ is infinite.*

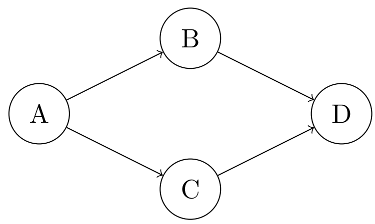
Exercise 3.3. Prove the preceding theorem.

In this section, we start to study multidimensional versions of the pigeonhole principle. To state them, we will need the notion of *graph*.

Intuitively, a graph is a pictorial representation of a network, i.e., a collection points, along with line segments connecting some of those points. The points represent the nodes of our network, and a line segment connecting two given points represents an interconnection of the network between the nodes represented by those two points. The interconnections between points of a graph may be directional, in which case the graph is said to be a *directed* graph, and the line segments are pictured as arrows. A directed graph represents flow: an arrow from point a to point b represents network flow from the node a to the node b (for example, if a and b represent cities, an arrow from a to b can be used to represent flow of information, flow of electricity, or flow of traffic from city a to city b). If a graph is non-directed (which is the default case), a line segment between point a and point b represents a nondirectional connection between a and b (for example, if a and b represent persons, a nondirectional edge between a and b can be used to represent that a and b are acquaintances, or that they attended the same high school). Historically, graph theory was considered to be a subarea of geometry; for this reason, the points of a graph $\mathscr{G}$ are historically referred to as the *vertices* of $\mathscr{G}$, and the line segments connecting points of $\mathscr{G}$ are called *edges* of $\mathscr{G}$.



Undirected Graph



Directed Graph

The formal definition of graph is as follows:

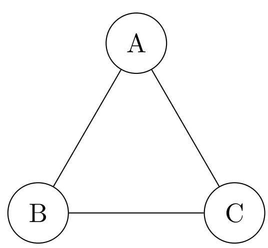
Definition 3.4. A *directed graph* is a structure of the form $\mathscr{G} = (G, R)$, where G is a set called the set of *vertices* (or *nodes*) of $\mathscr{G}$ and R is a binary relation on G called the *adjacency relation* of $\mathscr{G}$. The elements of the adjacency relation R are called the *edges* of $\mathscr{G}$. Thus, the edges of $\mathscr{G}$ connect vertices of $\mathscr{G}$. If $\mathscr{G} = (G, R)$ is a directed graph, two vertices a, b of $\mathscr{G}$ are *incident* if $a\ R\ b$, i.e., if they are connected by an edge of $\mathscr{G}$.

When the adjacency relation R is symmetric and antireflexive, we call $\mathscr{G}$ simply a *graph* (in other words, we drop the word "directed").

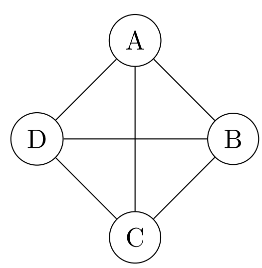
If $\mathscr{G} = (G, R)$ is a graph, a *subgraph* of $\mathscr{G}$ is any graph of the form $\mathscr{G}_0 = (G_0, R_0)$, where $G_0 \subseteq G$ and R_0 is the restriction of R to G_0.

A graph $\mathscr{G}$ is *complete* if any two vertices of $\mathscr{G}$ are incident (i.e., any two vertices of $\mathscr{G}$ are connected by an edge of $\mathscr{G}$). If n is a positive integer and $G = \{1, \ldots, n\}$, the complete graph $\mathscr{G} = (G, R)$ is denoted K_n. In other words, K_n is the complete graph on n points.

K_2



K_3



K_4

The graph $K_{\mathbb{N}}$ contains the integers $0, 1, 2, \ldots$ as vertices, with every pair (m, n) of distinct integers connected by and edge. Since the incidence relation is symmetric, it is useful to identify $K_{\mathbb{N}}$ with the set of all pairs of integers (m, n) satisfying $m < n$. This identification is standard in the mathematical literature.

$$\begin{bmatrix} \vdots & \vdots & \vdots & \vdots & \cdots \\ (0,4) & (1,4) & (2,4) & (3,4) & \\ (0,3) & (1,3) & (2,3) & & \\ (0,2) & (1,2) & & & \\ (0,1) & & & & \\ & & & & \end{bmatrix}$$

Definition 3.5. Given a graph $\mathscr{G} = (G, R)$, a *coloring* of $\mathscr{G}$ is a function f that to each edge of G assigns an element of a set C, which, in this context, we call a set of *colors*. If the set C has m elements, we say that f is a coloring of $\mathscr{G}$ into m colors. If f is a given coloring of $\mathscr{G}$, a subgraph $\mathscr{G}_0 = (G_0, R_0)$ of $\mathscr{G}$ is said to be *monochromatic* under f if the restriction of f to R_0 is constant, i.e., if f assigns the same color to all the edges of $\mathscr{G}_0$.

In general, a coloring of a set A into m colors, say, Color 1,..., Color m, is a function $f \colon A \to \{\text{Color } 1, \ldots, \text{Color } m\}$. This induces a partition of the set A into m mutually disjoint classes, namely, $f^{-1}(\text{Color } 1), \ldots, f^{-1}(\text{Color } m)$. Thus, any coloring function f, as above, partitions the set A into equivalence classes, and each "color" in the range of f provides a mathematical way to place a label on one of the classes.

Theorem 3.6 (Ramsey's theorem, classical version)**.** *For every pair of positive integers m, n, there exists a least positive integer $R_m(n)$ satisfying the following property: For every coloring f of the complete graph $K_{R_m(n)}$ into m many colors, there is a copy of K_n inside $K_{R_m(n)}$ that is monochromatic under f.*

Notation 3.7. To simplify notation, if the number m of colors is 2, we denote $R_m(n)$ simply as $R(n)$.

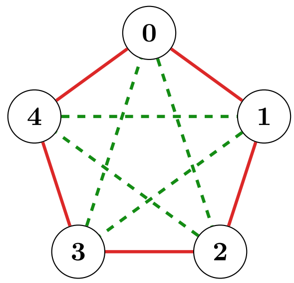
A popular interpretation of the existence of $R(n)$ is the following: Given any n, there exists $R(n)$ such that every party of $R(n)$ people contains a set of n individuals who are either mutual acquaintances or complete strangers.

Theorem 3.6, proved by F. P. Ramsey (not to be confused with Michael Ramsey, his brother, who was the Archbishop of Canterbury, or with Arthur Ramsey, his father, who was a prominent mathematician and president of Magdalene College) [**?**], has given rise to a very active line of research in mathematics known as *Ramsey theory* (or *partition theory*), in which the goal is to prove that if a given structure (not only a graph, but say, a group, a matrix algebra, or a Banach space) is sufficiently large, then we are guaranteed to find in it a substructure (respectively, a subgroup, a subalgebra, or a subspace) that is "organized" within the original structure in a recognizable way.

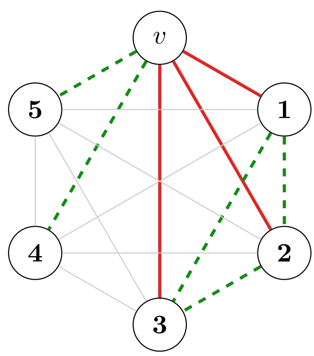
Examples 3.8.

(1) $R(2) = 2$. This is immediate from the definitions.

(2) $R(3) > 5$. To show this, it suffices to exhibit a coloring of K_5 into two colors, say Red and Green, that contains no all-Red triangle or all-Green triangle. To begin, organize the five vertices of K_5 as a pentagon, with all five sides of the pentagon colored Red, and every other edge of K_5 colored Green. It is immediate that this configuration contains no all-Red triangle or all-Green triangle.



(3) $R(3) = 6$. To show this, it suffices to show that every coloring of K_6 into colors, let's say, Red and Green, must contain an all-Red triangle or an all-Green triangle. To see this, organize the five vertices of K_5 as a hexagon and fix one of its vertices. Let's call this vertex v. By definition of K_5, there are 5 vertices incident to v. By the pigeonhole principle, three of them are the same color. Call these edges e_1, e_2, e_3, and, without loss of generality, assume that all of them are colored Red. We now consider two cases. If two of these red edges are connected by a red edge (Case 1), then we have found a red triangle. If not (Case 2), then the three vertices of e_1, e_2, e_3 not incident to v form a green triangle. In either case, we have a monochromatic triangle.



(4) $R(4) = 18$. The proof of this is similar to the preceding proof, but the number of cases to be considered is much larger.

(5) For $n > 5$, the Ramsey numbers $R(n)$ are increasingly computationally expensive to find. At the time of writing (fall semester, 2025), $R(5)$ is still unknown; all we know is that it lies between 43 and 46. The lower bound of 43 was obtained in 1989 by Geoffrey Exoo, and to this day, it is still the best [**?**]. An upper bound of 53 was obtained in 1992 by Brendan D. McKay and Stanislaw P. Radziszowski [**?**], and in a recent breakthrough (2024), it was improved by Vigleik Angelveit and Brendan McKay to 46 [**?**]. The proofs of these results

contain a heavy interplay between combinatorial graph theory and computational mathematics, so they fall outside the scope of our course.

3.2 Ramsey's Theorem (An Infinitary Version)

In order to prove Theorem 3.6, we will first prove a simple infinitary version that can be informally stated as follows: "Every party of infinitely many people contains infinitely many individuals who are either mutual acquaintances or complete strangers." The formal statement is below.

Theorem 3.9 (Infinitary version of Ramsey's theorem)**.** *Suppose that the set of all pairs (m,n) of positive integers with $m<n$ is partitioned into finitely many classes, or colors, say, Color 1, ..., Color k. Then there is an infinite increasing sequence*

$$n_0 < n_1 < n_2 < \cdots < n_k < \dots$$

of positive integers such that all the pairs of the form

$$(n_i, n_j), \qquad i < j$$

fall within the same class (color).

Before proving Theorem 3.9, let us observe an application to the theory of partially ordered sets:

Example 3.10. Let $(P, \leq)$ be a partially ordered set (see Definition 1.5). Two elements a, b of P are said to be *comparable* if $a \leq b$ or $b \leq a$. A subset $C \subseteq P$ is called a *chain* if any two elements of C are comparable, and an *antichain* if no two elements of C are comparable. By Theorem 3.9, every infinite partially ordered set contains either an infinite chain or an infinite antichain.

Proof of Theorem 3.9. Fix a coloring of all the pairs (m,n) for $m<n$ into finitely many colors. By the infinite pigeonhole principle (see 3.2), we can fix an increasing sequence

$$n_0^0 < n_1^0 < n_2^0 < \cdots < n_k^0 < \dots$$

of $(n)_{n\geq 0}$ such that the set of all pairs of the form

$$(n_0^0, n_j^0), \qquad j > 0$$

is monochromatic. Now we can find an increasing subsequence

$$n_1^1 < n_2^1 < n_3^1 < \cdots < n_k^1 < \dots$$

of $(n_j^0)_{j\geq 1}$ such that the set of all pairs of the form

$$(n_1^1, n_j^1), \qquad j > 1$$

is monochromatic. We continue this construction recursively: Given a subsequence

$$n_u^u < n_{u+1}^u < n_{u+2}^u < \cdots < n_k^u < \dots$$

of $(n_j^{u-1})_{j\geq u}$ such that each of the sequences

$$(n_0^0, n_j^0)_{j>0}, \dots, (n_{u-1}^{u-1}, n_j^{u-1})_{j>u-1}$$

are monochromatic, we construct a subsequence

$$n_{u+1}^{u+1} < n_{u+2}^{u+1} < n_{u+3}^{u+1} < \cdots < n_k^{u+1} < \dots$$

of $(n_j^u)_{j\geq u+1}$ such that each of the sequences

$$(n_0^0, n_j^0)_{j>0}, \ldots, (n_{u-1}^{u-1}, n_j^{u-1})_{j>u-1}, (n_u^u, n_j^u)_{j>u},$$

are monochromatic. Consider now the "diagonal" sequence $(n_u^*)_{u\in\mathbb{N}}$ defined by

$$n_u^* = n_u^u,$$

for all $u \in \mathbb{N}$. By construction, for each $u \in \mathbb{N}$, the sequence

$$(n_u^*, n_j^*)_{j>u} \tag{$*$}$$

is monochromatic.

Now, since the number of colors is finite and the number of possible $u \in \mathbb{N}$ above is infinite, we can invoke the infinite pigeonhole principle once again, and then, by replacing $(n_u^*)_{u\in\mathbb{N}}$ by an adequate subsequence if necessary, we can assume that all the subsequences in $(*)$ are the same color, regardless of $u \in \mathbb{N}$. This proves the theorem. $\square$

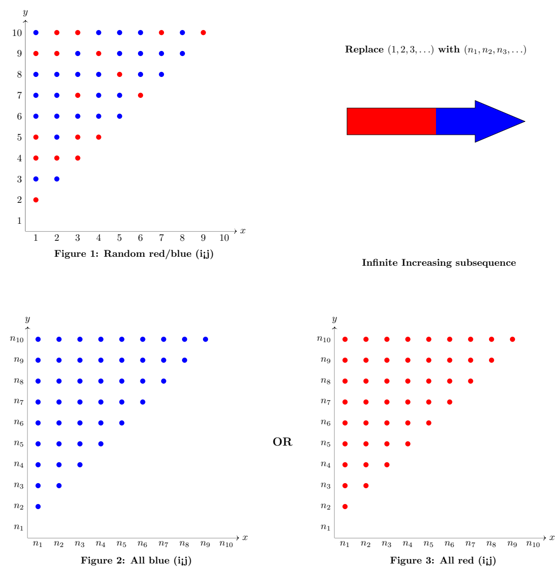


Figure 3.0. Illustration of the $(i < j)$ upper-triangle graphs with 2 colors.

Exercise 3.11. Prove that for every positive integer n there exists a positive integer N such that every partially ordered set of cardinality N contains either chains or antichains of length n.

Exercises 3.12.

(1) Prove that the version of Theorem 3.9 for finite colorings follows easily from the version for two colors. More precisely, prove that Theorem 3.9 as stated follows easily as a corollary from the case $k = 2$.

(2) We stated and proved Theorem 3.9 for pairs (m, n) with $m < n$. State a version of this theorem for finite l-tuples $(m_1, \ldots, m_l)$ of length $l \geq 2$, and prove that this apparently stronger version follows easily from the version for $l = 2$.

(3) Prove that the version of Theorem 3.6 for finite colorings follows as a corollary of the version for two colors. In other words, prove Theorem 3.6 from the particular case $m = 2$.

3.3 Real-Valued Relations and a Real-Valued Version of Ramsey's Theorem

Definition 3.13. Let A be a set and n a nonnegative integer. A $[0,1]$-*valued* n-*ary relation* on A is a function $R\colon A^n \to [0,1]$.

Theorem 3.14 (Infinitary version of Ramsey's theorem)**.** *Suppose that* f *is a function that assigns to every pair* (m,n) *of positive integers with* $m < n$ *a "color"* $f(m,n) \in [0,1]$*. Then there exists a sequence*

$$n_0 < n_1 < n_2 < \cdots < n_k < \dots$$

and a "limit color" $c \in [0,1]$ *with the property that*

$$\lim_{\substack{i,j\to\infty \\ i<j}} f(n_i, n_j) = c.$$

The latter means that for every $\epsilon > 0$ *there exists* $N = N(\epsilon) > 0$ *such that*

$$\forall i,j \; (\, N < i < j \quad \Longrightarrow \quad f(n_i,n_j) \in (c-\epsilon, c+\epsilon)\,).$$

Proof. Fix f as in the statement of Theorem 3.14. Split the interval $[0,1]$ as a union of two closed subintervals, say, $[0,1] = I \cup J$, with I and J of length $\frac{1}{2}$. By Theorem 3.9, there exists a sequence

$$n_0^0 < n_1^0 < n_2^0 < \cdots < n_k^0 < \dots$$

such that $f(n_i^0, n_j^0) \in I$ for all $i < j$, or $f(n_i^0, n_j^0) \in J$ for all $i < j$. In the former case, let $I_0 = I$ and in the latter case, let $I_0 = J$. Now we continue this process recursively: For $u = 1, 2, \dots$, we construct sequences

$$n_0^u < n_1^u < n_2^u < \cdots < n_k^u < \dots$$

and subintervals I_u of the unit interval such that for all $u \in \mathbb{N}$,

(1) $(n_j^{u+1})_{j\in\mathbb{N}}$ is a subsequence of $(n_j^u)_{j\in\mathbb{N}}$,

(2) I_{u+1} is a subinterval of I_u of length $\frac{1}{2^{u+1}}$,

(3) $f(n_i^u, n_j^u) \in I_u$ for all $i < j$.

Let $c \in [0,1]$ be such that

$$c \in \bigcap_{u\in\mathbb{N}} I_u,$$

and for each $u \in \mathbb{N}$, define

$$n_k = n_k^k.$$

Then, $(n_k)_{k\in\mathbb{N}}$ has the desired property. ☐

Notice that in the preceding proof, the interval $[0,1]$ can be replaced by any closed interval $[a,b]$.

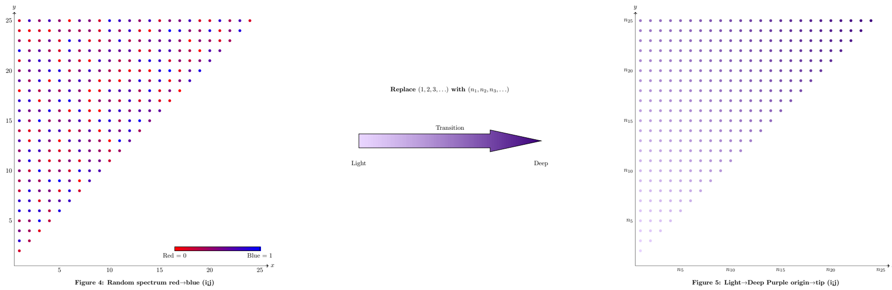


Figure 3.1. Illustration of the $(i < j)$ upper-triangle graphs with a spectrum.

Exercises 3.15.

(1) Show through a counterexample that Theorem 3.14 is false if the interval $[0, 1]$ is replaced by $\mathbb{R}$.

(2) Let $[a_0, b_0], [a_1, b_1], \ldots, [a_k, b_k], \ldots$ be a countable sequence of closed intervals, and suppose that f is a function that assigns to every pair (m, n) of positive integers with $m < n$ a "color" $f(m, n) \in \prod_{u\in\mathbb{N}}[a_u, b_u]$. Prove that there exists a sequence

$$n_0 < n_1 < n_2 < \cdots < n_k < \ldots$$

and a "limit color"

$$(c_u) \in \prod_{u\in\mathbb{N}} [a_u, b_u]$$

with the following property: For every $\epsilon > 0$ and every $k \in \mathbb{N}$, there exists $N = N(\epsilon, k) > 0$ such that

$$\forall i, j\ (\,N < i < j \quad\Longrightarrow\quad f(n_i, n_j) \in (c_u - \epsilon, c_u + \epsilon)\,),$$

for every $u \leq k$.

4

Limits of structures

Recall from 2.12 that if $\mathscr{F}$ is a filter on a nonempty set I, then the elements of $\mathscr{F}$ are called *$\mathscr{F}$-large* subsets of I. A property of elements of I is said to hold for *almost every $i \in I$* or *almost everywhere* in $I \pmod{\mathscr{F}}$ if the property holds for an $\mathscr{F}$-large subset of I.

Recall also that if $(G_i)_{i\in I}$ is an indexed family of sets, then the *cartesian product* of the family $(G_i)_{i\in I}$, denoted $\prod_{i\in I} G_i$, is defined as

$$\prod_{i\in I} G_i = \{(a_i)_{i\in I} \mid a_i \in G_i \text{ for all } i \in I\}.$$

Fix an indexed family $(G_i)_{i\in I}$ of sets and an ultrafilter $\mathscr{U}$ as above. For any elements $a = (a_i)_{i\in I}$ and $b = (b_i)_{i\in I}$ in the cartesian product $\prod_{i\in I} G_i$, define a binary relation $\sim$ by

$$a \sim b \quad \text{if and only if} \quad a_i = b_i \text{ for almost every } i \in I \pmod{\mathscr{U}},$$

i.e.,

$$a \sim b \quad \text{if and only if} \quad \{i \in I\colon a_i = b_i\} \in \mathscr{U}.$$

Exercise 4.1. (a) Prove that $\sim$ is an equivalence relation on the set I.

(b) Do we need for $\mathscr{U}$ to be an ultrafilter, or does a filter suffice for $\sim$ to be an equivalence relation?

Definition 4.2. If $(G_i)_{i\in I}$ is an indexed family of sets and $\mathscr{U}$ is an ultrafilter on the index set I, then the set

$$\prod_{i\in I} G_i/\sim$$

of equivalence classes of $\prod_{i\in I} G_i$ under the equivalence relation $\sim$ is called the *$\mathscr{U}$-ultraproduct* of the family $(G_i)_{i\in I}$. It is denoted

$$\prod_{\mathscr{U}} G_i$$

4.1 Graph limits

Fix an indexed family $(\mathscr{G}_i)_{i\in I}$ of graphs, where $\mathscr{G}_i = (G_i, R_i)$ for all $i \in I$, and an ultrafilter filter $\mathscr{U}$. For any elements $a = (a_i)_{i\in I}$ and $b = (b_i)_{i\in I}$ in the cartesian product $\prod_{i\in I} G_i$, define a binary relation $R_{\mathscr{U}}$ by

$$aR_{\mathscr{U}}b \quad \text{if and only if} \quad a_iR_ib_i \text{ for almost every } i \in I \pmod{\mathscr{U}}.$$

Exercise 4.3. Prove that the relation $R_{\mathscr{U}}$ defined above is well-defined, that is, if $aR_{\mathscr{U}}b$ and $a \sim a', b \sim b'$, then $a'R_{\mathscr{U}}b'$.

The following is the most important definition of this chapter, and one of the central definitions of the course.

Definition 4.4. If $(\mathscr{G}_i)_{i\in I}$ is an indexed family of graphs, where $\mathscr{G}_i = (G_i, R_i)$ for all $i \in I$, and $\mathscr{U}$ is an ultrafilter on the index set I, then the graph

$$\left(\prod_{\mathscr{U}} G_i, R_{\mathscr{U}}\right)$$

is called the *$\mathscr{U}$-limit* of the family $(\mathscr{G}_i)_{i\in I}$. It is denoted

$$\lim_{\mathscr{U}} \mathscr{G}_i.$$

4.2 The Transfer Principle for graphs

Let $x, x', x'' \ldots, y, y', y'', \ldots, z, z', z'' \ldots$ be a fixed countable set of variables. An *atomic expression* is any expression of the form

$$xRy$$

or

$$x = y,$$

where x and y are any of the variables from our fixed list.

Definition 4.5. Starting with the atomic expressions, we can form more complex expressions, known as *first-order formulas*, by any finite number of applications of the following recursive rules:

(1) Any atomic expression is a first-order formula.

(2) If α and β are first-order formulas, then the expressions

- $\neg\alpha$
- $(\alpha \wedge \beta)$
- $(\alpha \vee \beta)$

are first-order formulas. Note that we don't need $(\alpha \vee \beta)$ since we can regard it as an abbreviation of $\neg(\neg\alpha \vee \neg\beta)$. Similarly, we regard $(\alpha \to \beta)$ as an abbreviation of $(\neg\alpha \vee \beta)$.

(3) If α is a first-order formula and x is a variable, then the expressions

- $\forall x\alpha$
- $\exists x\alpha$

are first-order formulas. Note that we don't need $\exists x\alpha$ since we can regard it as an abbreviation of $\neg\forall x\neg\alpha$.

Note that a first-order formula is a of finite length, and therefore it contains only finitely many variables.

Definition 4.6. If α is a first-order formula such that all the variables of α are under the scope of a quantifier ($\forall$ or $\exists$), we say that α is a *sentence*.

Remark 4.7. Note that if α is a sentence and $\mathscr{G}$ is any graph, then either α is true in $\mathscr{G}$ or α is false in $\mathscr{G}$.

Notation 4.8. If α is true in $\mathscr{G}$, we write

$$G \models \alpha.$$

Examples 4.9.

(1) If (G, R) is a graph, then R is a linear order on G if and only if

$$\begin{aligned}(G, R) \models & \forall x(xRx)\wedge \\ & \forall x\forall y(xRy \wedge yRx \to x = y)\wedge \\ & \forall x\forall y\forall z(xRy \wedge yRz \to xRz)\wedge \\ & \forall x\forall y(xRy \vee yRx).\end{aligned}$$

(2) If (G, R) is a graph, (G, R) contains a partial order of 50 elements if and only if $(G, R) \models \alpha$, where α is the following first-order sentence:

$$\exists x_1\exists x_2\cdots\exists x_{50}\Bigg[\underbrace{\bigwedge_{i<j\leq 50} \neg(x_i = x_j)}_{x_1,\ldots,x_{50}\text{ are distinct}} \wedge \underbrace{\forall x\Big(\big(\bigvee_{i\leq 50} x = x_i\big) \to xRx\Big)}_{\text{reflexivity on the set }\{x_1,\ldots,x_{50}\}} \wedge$$

$$\underbrace{\forall x\forall y\Big(\big(\bigvee_{i\leq 50} x = x_i\big) \wedge \big(\bigvee_{i\leq 50} y = x_i\big) \to \big((xRy \wedge yRx) \to x = y\big)\Big)}_{\text{antisymmetry}} \wedge$$

$$\underbrace{\forall x\forall y\forall z\Big(\big(\bigvee_{i\leq 50} x = x_i\big) \wedge \big(\bigvee_{i\leq 50} y = x_i\big) \wedge \big(\bigvee_{i\leq 50} z = x_i\big) \to \big((xRy \wedge yRz) \to xRz\big)\Big)}_{\text{transitivity}}\Bigg].$$

(3) If (G, R) is a graph and $n \in \mathbb{N}$, then G has a monochromatic subset of n elements if and only if $(G, R) \models \alpha$, where α is the following first-order sentence:

$$\exists x_1\exists x_2\cdots\exists x_n\Bigg[\underbrace{\bigwedge_{i<j\leq n} \neg(x_i = x_j)}_{x_1,\ldots,x_n\text{ are distinct}} \wedge \underbrace{\forall x\forall y\Big(\big(\bigvee_{i<j\leq n} x = x_i\big) \wedge \big(y = x_j\big) \to R(x, y)\Big)}_{\text{all the pairs in }\{x_1,\ldots,x_n\}\text{ are incident}} \vee$$

$$\underbrace{\forall x\forall y\Big(\big(\bigvee_{i<j\leq n} x = x_i\big) \wedge \big(y = x_j\big) \to \neg R(x, y)\Big)}_{\text{none of the pairs in }\{x_1,\ldots,x_n\}\text{ are incident}}\Bigg].$$

Theorem 4.10 (The Transfer Principle for graph limits). *Let $(\mathscr{G}_i)_{i\in I}$ be an indexed family of graphs, let $\mathscr{U}$ be an ultrafilter on the index set I, and let $\lim_{\mathscr{U}} \mathscr{G}_i$ denote the $\mathscr{U}$-limit of $(\mathscr{G}_i)_{i\in I}$ (see Definition 4.4). Then, for any first-order sentence α, we have*

$$\lim_{\mathscr{U}} \mathscr{G}_i \models \alpha \quad \textit{if and only if} \qquad \mathscr{G}_i \models \alpha \textit{ for almost every } i \in I \pmod{\mathscr{U}}.$$

We will prove the Transfer Principle as a corollary of a stronger theorem (see Theorem 4.14). However, before engaging in the proof of the theorem, let us focus on an important application.

4.2.1 Back to Ramsey's theorem. In Chapter 3, we proved the infinitary version of Ramsey's theorem (see 3.9). However, we did not prove the classical finitary version 3.6. We now show that the Transfer Principle for graph limits allows us to obtain the finitary version in a few lines from the infinitary one.

Theorem 4.11 (Ramsey's theorem). *For every $n \in \mathbb{N}$, there exists a positive integer $R(n)$ such that if we color $K_{R(n)}$ into two colors, say "black" and "white", then there is a subgraph of $K_{R(n)}$ isomorphic to K_n such that either all edges of this smaller graph are black, or all edges of this graph are white.*

Proof. Assume that the theorem does not hold true, and fix $n \in \mathbb{N}$ for which $R(n)$ doesn't exist. This means that, for each $N \in \mathbb{N}$ we can fix a a graph $\mathscr{G}_N = (G_N, R_N)$ of size $> N$ that has no monochromatic subset of size n. Fix a non-principal ultrafilter $\mathscr{U}$ on $\mathbb{N}$. Let α be the sentence of Examples that asserts the existence of a monochromatic subgraph of size n. Then $\mathscr{G}_N \models \neg\alpha$. Let $\lim_{\mathscr{U}} \mathscr{G}_N$ denote the $\mathscr{U}$-limit of $(\mathscr{G}_N)_{N\in\mathbb{N}}$. Then, by the Transfer Principle for graph limits, $\lim_{\mathscr{U}} \mathscr{G}_i \models \neg\alpha$. But this contradicts the infinitary version of Ramsey's theorem (see 3.9) since $\lim_{\mathscr{U}} \mathscr{G}_N$ is an infinite graph. $\square$

Exercise 4.12. For every $m, n \in \mathbb{N}$, there exists $R(m, n)$ such that if we color $K_{R(m,n)}$ into two colors, "black" and "white", then there is a subgraph of $K_{R(m,n)}$ isomorphic to K_m all of whose edges are black, or subgraph of $K_{R(m,n)}$ isomorphic to K_n all of whose edges are white.

4.3 General statement and proof of the Transfer Principle for graph limits

Recall (see Definition **??**) that a sentence is a first-order formula where all the variables are under the scope of a quantifier. If α is not a sentence, then those variables in α that are not under the scope of a quantifier are said to occur *free* in α. If all the free variables that occur in α are in the set $\{x_1, \ldots, x_n\}$, we write α as $\alpha(x_1, \ldots, x_n)$. For example, if α is the formula $\exists y(xRy)$, then we may write α as $\alpha(x)$.

As we indicated in Remark **??**, if α is a sentence and $\mathscr{G}$ is any graph, then either $\mathscr{G} \models \alpha$ or $\mathscr{G} \not\models \alpha$. However, if α is not a sentence, i.e., we cannot say whether or not $\mathscr{G} \models \alpha$. To see an example of this, consider the formula $\alpha(x)$ defined as $\exists y(xRy)$ and the graph $\mathscr{G} = (G, R)$ where $G = \{a, b, c\}$ and $R = (a, b)$. Then, we could say that $\mathscr{G} \models \alpha$ if we interpret the variable x as a, and $\mathscr{G} \not\models \alpha$ if we interpret the variable x as c. In general, if α includes free variables that are free, to determine whether $G \models \alpha$, we need to interpret the free occurrences of free variables in α as elements of $\mathscr{G}$.

Notation 4.13. Let $\alpha(x_1, \dots, x_n)$ be a first-order formula with free variables in the set $\{x_1, \dots, x_n\}$ and let $\mathscr{G} = (G, R)$ be a graph. If $\alpha_1, \dots, a_n$ are elements of G, we write

$$\mathscr{G} \models \alpha[a_1, \dots, a_n]$$

If $\alpha(x_1, \dots, x_n)$ is true in $\mathscr{G}$ when x_k is interpreted as a_k, for $k = 1, \dots, n$.

Now we can state and prove the Transfer Principle for graph limits.

Theorem 4.14 (The Transfer Principle for graphs)**.** *Let $(\mathscr{G}_i)_{i\in I}$ be an indexed family of graphs, let $\mathscr{U}$ be an ultrafilter on the index set I, and let $\lim_{\mathscr{U}} \mathscr{G}_i$ denote the $\mathscr{U}$-limit of $(\mathscr{G}_i)_{i\in I}$. Then, if $\alpha(x_1, \dots, x_n)$ is a first-order sentence and $a_1, \dots, a_n$ are elements of $\lim_{\mathscr{U}} \mathscr{G}_i$, represented by families $a_k = (a_k(i))_{i\in I}$ for $k = 1, \dots, n$, then*

$$\begin{gathered} \lim_{\mathscr{U}} \mathscr{G}_i \models \alpha[a_1, \dots, a_n] \\ \textit{if and only if} \\ \mathscr{G}_i \models \alpha[a_1(i), \dots, a_n(i)] \textit{ for almost every } i \in I \pmod{\mathscr{U}}. \end{gathered} \tag{*}$$

In the proof, we will invoke the following exercise:

Exercise 4.15. Let $\mathscr{U}$ be an ultrafilter on I, an indexed set. If $A, B \subseteq I$, then

(1) $A \cap B \in \mathscr{U}$ if and only if $A \in \mathscr{U}$ and $B \in \mathscr{U}$.

(2) $A \cup B \in \mathscr{U}$ if and only if $A \in \mathscr{U}$ or $B \in \mathscr{U}$.

(3) $A^{\mathsf{c}} \in \mathscr{U}$ if and only if $A \notin \mathscr{U}$.

Proof of Theorem 4.14. We prove the equivalence $(*)$ by induction on the complexity of α. When α is an atomic formula, the equivalence holds by the very definition of $\lim_{\mathscr{U}} \mathscr{G}_i$ (see Definition 4.4). We now assume, by induction, that the equivalence $(*)$ holds for α and β, and we prove it for $(\alpha \wedge \beta)$ and $\neg\alpha$.

First, we suppose, as induction hypothesis, that $(*)$ holds for $\alpha(x_1, \dots, x_n)$ and $\beta(x_1, \dots, x_n)$ in order to prove it for $(\alpha \wedge \beta)(x_1, \dots, x_n)$. Let

$$\begin{aligned} A &= \{i \in I : \mathscr{G}_i \models \alpha[a_1(i), \dots, a_n(i)]\}, \\ B &= \{i \in I : \mathscr{G}_i \models \beta[a_1(i), \dots, a_n(i)]\}. \end{aligned}$$

It follows immediately from the definition of $\models$ that

$$A \cap B = \{i \in I : \mathscr{G}_i \models (\alpha \wedge \beta)[a_1(i), \dots, a_n(i)]\}.$$

Hence,

$$\begin{array}{lll}
& \lim\limits_{\mathscr{U}} \mathscr{G}_i \models (\alpha \wedge \beta)[a_1, \ldots, a_n] & \\
\text{iff} & \lim\limits_{\mathscr{U}} \mathscr{G}_i \models \alpha[a_1, \ldots, a_n] \text{ and } \lim\limits_{\mathscr{U}} \mathscr{G}_i \models \beta[a_1, \ldots, a_n] & \text{by definition of } \models \\
\text{iff} & A, B \in \mathscr{U} & \text{by induction hypothesis} \\
\text{iff} & A \cap B \in \mathscr{U} & \text{by Exercise 4.15} \\
\text{iff} & \mathscr{G}_i \models (\alpha \wedge \beta)[a_1(i), \ldots, a_n(i)], \text{ for almost every } i \in I \pmod{\mathscr{U}}. &
\end{array}$$

The proof for the case $\neg\alpha(x_1, \ldots, x_n)$ is similar. Suppose that $(*)$ holds for $\alpha(x_1, \ldots, x_n)$. Evidently,

$$A^{\mathrm{c}} = \{i \in I \colon \mathscr{G}_i \not\models \alpha[a_1(i), \ldots, a_n(i)]\}.$$

Hence,

$$\begin{array}{lll}
& \lim\limits_{\mathscr{U}} \mathscr{G}_i \models \neg\alpha[a_1, \ldots, a_n] & \\
\text{iff} & \lim\limits_{\mathscr{U}} \mathscr{G}_i \not\models \alpha[a_1, \ldots, a_n] & \text{by definition of } \models \\
\text{iff} & A \notin \mathscr{U} & \text{by induction hypothesis} \\
\text{iff} & A^c \in \mathscr{U} & \text{by Exercise 4.15} \\
\text{iff} & \mathscr{G}_i \models \neg\alpha[a_1(i), \ldots, a_n(i)], \text{ for almost every } i \in I \pmod{\mathscr{U}}. &
\end{array}$$

The case when α is of the form $\exists y \beta(x_1, \ldots, x_n y)$ is left as an exercise. $\square$

Exercise 4.16. Finish the proof of Theorem 4.14. [Advice: Treat each of the two directions separately.]

Exercise 4.17. (a) Exhibit a sequence $(G_n, R_n)_{n \in \mathbb{N}}$ of graphs and an ultrafilter $\mathscr{U}$ such that (G_n, R_n) is a well-ordered set for each $n \in \mathbb{N}$, but $\lim_{\mathscr{U}}(G_n, R_n)$ is not well-ordered.

(b) Can you write a first-order sentence α such that $(G, R) \models \alpha$ if and only if (G, R) is a well-ordered set?